



Document Management System – Project README



Project Overview

Project Name: (Add your system name here)

The **Document Management System (DMS)** is a centralized platform designed to manage PDFs and multiple document file formats efficiently in one place. The platform enables users to organize documents through **channels**, collaborate with members, and securely control access using role-based permissions.

Unlike messaging platforms such as WhatsApp or Telegram, this system is **purpose-built specifically for structured document storage, access control, and collaboration**.



Objectives

- Centralized document storage
 - Channel-based document organization
 - Role-based access control (Admin / Member)
 - Secure sharing through unique channel codes
 - Lightweight collaboration via document reactions
 - Simple and scalable interface for academic and professional usage
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User Roles



Channel Admin (Creator)

Permissions: - Create channels - Upload documents - View all members - Assign admin rights to members - Control number of allowed members



Channel Member

Permissions: - View documents - React to documents - Access joined channels



Authentication Flow

Users must register and login before accessing the system.

Registration

User provides: - Username - Email - Password

System creates account and redirects to login.

Login

User enters: - Email - Password

System validates credentials and redirects to Dashboard.



Dashboard Overview

After login, user sees **three main sections**:

1. My Channels
2. Search Channels
3. Create Channel



Feature Breakdown



My Channels

Displays: - All channels joined by the user

User can: - Open channel - View documents - React to documents

Admin can additionally: - Upload documents - View members list - Promote members to admin



Search Channels

Users can: - Search channel using **Unique Channel Code** - Join channel (if member limit not exceeded)



Create Channel

Admin provides: - Channel Name - Optional Member Limit

System generates: - Unique Channel Code

Channel gets created successfully.

Channel Workspace Features

Inside each channel:

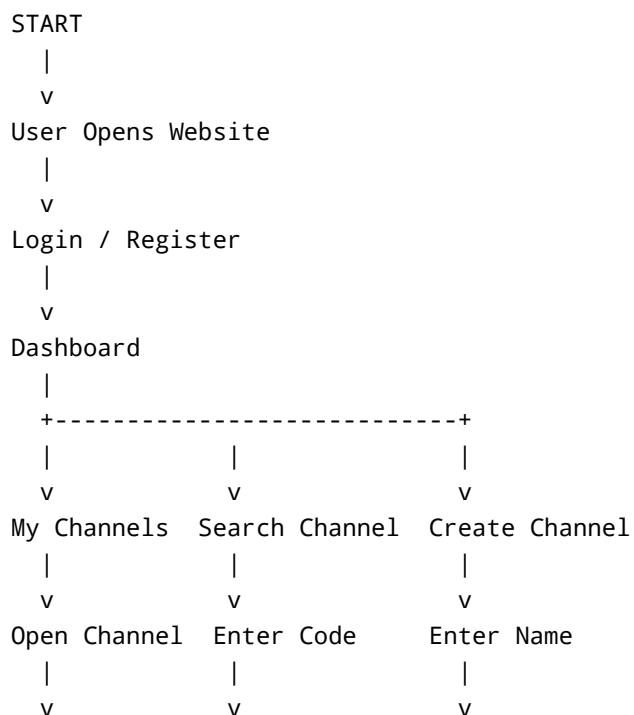
Admin Actions: - Upload PDFs - Upload document files - Manage members - Promote members to admin

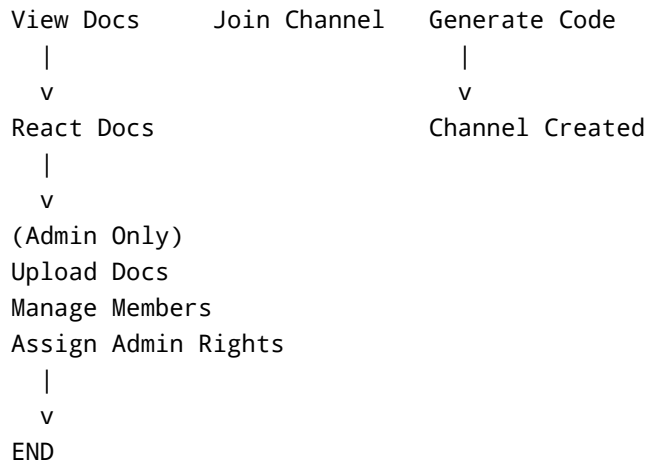
Member Actions: - View documents - React to documents

Supported file types:

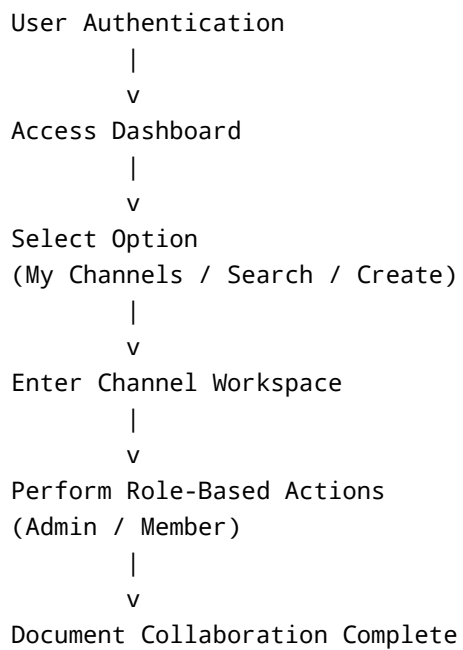
PDF
DOC
DOCX
TXT
PPT
XLS
and more

Complete User Flow Diagram





System Workflow (Detailed)



Security Features

- Secure login authentication
- Channel access via unique channel code
- Role-based permission system
- Optional member limit restriction



Future Enhancements

Planned upgrades: - Version control for documents - Document tagging system - Download permission control - Activity logs - Search inside documents - Notifications system - Cloud storage integration



Tech Stack

Frontend: - React.js - TailwindCSS

Backend: - Python (Flask)

Database: - MySQL

Authentication: - Custom made token system

Storage: - Local server / Cloud storage



Use Cases

This system is ideal for:

- Students managing subject materials
 - Teachers sharing academic resources
 - Teams managing project documentation
 - Organizations handling structured file sharing
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Summary

The **Document Management System** provides a structured alternative to chat-based file sharing platforms by focusing purely on **organized document collaboration through channel-based architecture**.

It ensures clarity, security, and scalability for academic and professional workflows.



Author

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